Statistical Modeling

DATA.STAT.740
Lecture 11

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Penalized regression

In penalized regression model we add a penalty term to the optimization problem. Penalized methods are applicable to GLMs but we use it only to normal linear model as an example

Ordinary least squares fitting: $\hat{\beta} = \arg\min_{\beta} \sum_{i=1}^{N} (y_i - x_i^T \beta)^2$

Penalized regression: $\hat{\beta} = \arg\min_{\beta} \sum_{i=1}^{N} (y_i - x_i^T \beta)^2 + \lambda \cdot \text{penalty}(\beta)$

In penalized regression, almost always the predictors are centered and scaled this way all the coefficients are penalized fairly

Penalized regression

Reasons for using penalized regression

1. overfitting
   The model explains the noise and is not usable for predictions

2. multicollinearity
   When the predictors are correlated, the estimation may become unstable

3. High dimensionality
   If there are more predictors than data, the model can’t be fit without penalization - (also called regularization)

4. Variance reduction and more stable model prediction
   Penalization shifts coefficients towards zero and reduces variance

Common penalization terms

$$\hat{\beta} = \arg\min_{\beta} \sum_{i=1}^{N} (y_i - x_i^T \beta)^2 + \lambda \cdot \text{penalty}(\beta)$$

L2 penalty : Ridge regression

$$\text{penalty}(\beta) = \sum_{j} \beta_j^2$$

Shrinks coefficients smoothly toward zero

Never sets the coefficients exactly to zero

Good for multicollinearity

Closed-form solution exists $\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$

Common penalization terms

$$\hat{\beta} = \arg\min_{\beta} \sum_{i=1}^{N} (y_i - x_i^T \beta)^2 + \lambda \cdot \text{penalty}(\beta)$$

L1 penalty : LASSO regression

$$\text{penalty}(\beta) = \sum_{j} |\beta_j|$$

Shrinks coefficients exactly to zero

Can be used for variable selection

Useful when you have many predictors and some one them are irrelevant

No closed form solution but there are efficient numerical solutions

Common penalization terms

$$\hat{\beta} = \arg\min_{\beta} \sum_{i=1}^{N} (y_i - x_i^T \beta)^2 + \lambda \cdot \text{penalty}(\beta)$$

L1 + L2 penalty combined : elastic net regression

$$\text{penalty}(\beta) = \alpha \sum_j |\beta_j| + (1-\alpha) \sum_j \beta_j^2$$

Mix of LASSO and ridge regression

Aim is to combine stability of ridge and sparsity of LASSO

Ridge regression

Consider the linear normal model $\mathbf{y} = X\beta + \boldsymbol{\epsilon},\ \boldsymbol{\epsilon} \sim \text{Normal}(0, \sigma^2 I)$

If the columns (the predictors) of the design matrix are highly correlated, the traditional maximum likelihood estimate may be very unstable

In the case that the rank of the matrix is less than the number of predictors then estimate $\hat{\beta} = (X^T X)^- X^T y$ still exists with $(X^T X)^-$ being the generalized inverse - but **the solution is not unique**

Ridge regression

A ridge estimator can be used to modify the problem so that from the nonunique set of estimates we select the one “closest to origin”

The modified model is $\begin{bmatrix} \mathbf{y} \\ 0 \end{bmatrix} = \begin{bmatrix} X \\ \sqrt{\lambda} I \end{bmatrix} \beta + \begin{bmatrix} \boldsymbol{\epsilon} \\ \boldsymbol{\epsilon}_R \end{bmatrix}$

The “fake” observation i.e. the stochastic restriction $0 = \sqrt{\lambda}\beta + \boldsymbol{\epsilon}_R$ will shrink the product towards the matrix 0

if λ is large, then the ridge regression estimate needs to shrink towards 0
The ridge estimator is also referred to as the shrinkage estimator

Problem can also be formulated as $\hat{\beta} = \arg\min_{\beta} (y - X\beta)^T (y - X\beta) + \lambda \cdot \beta^T \beta$

Ridge regression

The ridge estimator $\hat{\boldsymbol{\beta}} = (X^T X + \lambda I)^{-1} X^T \mathbf{y}$ is a biased estimator:

$$\mathsf{E}(\hat{\boldsymbol{\beta}}) = (X^T X + \lambda I)^{-1} X^T \mathsf{E}(\mathbf{y}) = (X^T X + \lambda I)^{-1} X^T \beta$$

and the estimator covariance is

$$\mathsf{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (X^T X + \lambda I)^{-1} X^T X (X^T X + \lambda I)^{-1}$$

Often no penalty is assigned to the intercept term but only to the slopes and we deal with centered and scaled predictors

$$\mathbf{y} = \mathbf{1}\beta_0 + X_1 \beta_1 + \boldsymbol{\epsilon}$$

$$\min_{\beta_0, \beta_1} (\mathbf{y} - \mathbf{1}\beta_0 - X_1\beta_1)^T (\mathbf{y} - \mathbf{1}\beta_0 - X_1\beta_1) + \lambda \beta_1^T \beta_1$$

Ridge regression

The partitioned model $\mathbf{y} = \mathbf{1}\beta_0 + X_1\beta_1 + \boldsymbol{\epsilon}$

$$\min_{\beta_0,\beta_1} (\mathbf{y} - \mathbf{1}\beta_0 - X_1\beta_1)^T(\mathbf{y} - \mathbf{1}\beta_0 - X_1\beta_1) + \lambda\beta_1^T\beta_1$$

Now the penalty term doesn't influence the intercept term and its estimate is

Center $\quad y^{(c)} = y - \bar{y}, \quad X^{(c)} = X_1 - \mathbf{1}\bar{x}_1^T \quad$ where $\quad \bar{y} = \frac{1}{N}\mathbf{1}^T y$

Scale $\quad X^{(cs)} = X^{(c)} D^{-1}$

Estimation $\quad \hat{\beta}^{(cs)} = ({X^{(cs)}}^T X^{(cs)} + \lambda)^{-1} {X^{(cs)}}^T y^c$

$$\bar{x}_1 = \frac{1}{N}\mathbf{1}^T X_1$$

Back-transform $\quad \hat{\beta} = D^{-1}\hat{\beta}^{(cs)}, \quad \hat{\beta}_0 = \bar{y} - \bar{x}^T\hat{\beta}$

$$[D]_{i,i} = \mathsf{sd}(x_i)$$

LASSO regression

The interest is to select important explanatory variables or do feature selection for a large group of possible explanatory variables

Use LASSO estimation method for automatic selection

Can also be used if there are more plausible predictors than there is data

Consider the partitioned linear model where the predictors have been centered and scaled

$$\mathbf{y} = \mathbf{1}\beta_0 + X_1\boldsymbol{\beta}_1 + \boldsymbol{\epsilon},\ \boldsymbol{\epsilon} \sim \textsf{Normal}(0, \sigma^2 I),\ \boldsymbol{\beta}_1 = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}$$

LASSO regression

The lasso estimator is obtained as a solution to minimization problem

$$\min_{\beta_0, \boldsymbol{\beta}_1} (\mathbf{y} - \mathbf{1}\beta_0 - X_1\boldsymbol{\beta}_1)^T (\mathbf{y} - \mathbf{1}\beta_0 - X_1\boldsymbol{\beta}_1) + \lambda \sum_{k=1}^{p} |\beta_k|$$

When using L2 penalty in ridge regression, the coefficients never go to zero, but with L1 penalty in LASSO

$$\lambda \sum_{k=1}^{p} |\beta_k|$$

increasing λ will cause some of the coefficients to collapse to 0

LASSO model can't be fitted analytically

R glmnet

glmnet-package in R is a popular package for GLMs penalties

$$\min_{\beta_0,\boldsymbol{\beta}_1} \frac{1}{2N}(\mathbf{y} - \mathbf{1}\beta_0 - X_1\boldsymbol{\beta}_1)^T(\mathbf{y} - \mathbf{1}\beta_0 - X_1\boldsymbol{\beta}_1) + \lambda \cdot \left[(1-\alpha)\cdot\frac{\boldsymbol{\beta}_1^T\boldsymbol{\beta}_1}{2} + \alpha \cdot \sum_{k=1}^{p}|\beta_k|\right]$$

$\alpha = 0$: ridge

$\alpha = 1$: LASSO

$0 < \alpha < 1$: elastic net

The tuning parameter λ controls the strength of the penalty

Can be estimated (automatically) using k-fold cross-validation

1) Hold of 1/k part of the data
2) Fit a model with rest of the data
3) Calculate the prediction error of the model to the unused data
4) Go back to 1) with another 1/k part of the data

R-example!

Survival analysis

Statistics for analyzing time-to-event data
- We record
  - A starting point
  - The time until an event occurs
  - An ending point (end of a follow-up)

Event-times are often highly skewed and we can’t model it with ordinary regression

Survival analysis

An event is not always recorded during the time period we observe a statistical unit

The data is then **censored**

Either the statistical unit dropped out (censored at dropout time)

or an event has not yet happened - yet (right-censored data)

The **risk** of an event happening may change over time

Survival analysis provides the tools for analyzing time-to-event data and handling censoring and modeling time-dependent risk

We say that a statistical unit **is at risk** at time t if

They have not yet had the event & They have not been censored before t

Data example

Let the study period be [0, 15]

Patient	Survival Time	Event occurred?
1	10	Yes
2	8	Yes
3	15	No

This is a typical time-to-event dataset

We have observation times and an indicator variable whether event occurred

If event didn't occur at the end of the study period, then the data is censored

Survival analysis

Survival function - probability of the event not happening before t

$$S(t) = \mathsf{P}(T > t)$$

Probability density function

$$f(t) = \frac{d}{dt}(1 - S(t))$$

Hazard function - the risk of event happening at time $(t, t + \Delta t)$ having survived until time t

$$h(t) = \lim_{\Delta t \to 0} \frac{\mathsf{P}(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t}$$

Increasing: e.g. aging

Decreasing: e.g. infant mortality

Constant: memoryless

Hazard function

Hazard function can be expressed as

$$\begin{aligned} h(t) &= \lim_{\Delta t \to 0} \frac{\mathsf{P}(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t} \\ &= \lim_{\Delta t \to 0} \frac{\mathsf{P}(t \leq T < t + \Delta t)}{\mathsf{P}(T \geq t)\Delta t} \\ &= \lim_{\Delta t \to 0} \frac{\mathsf{P}(t \leq T < t + \Delta t)}{S(t)\Delta t} = \frac{f(t)}{S(t)} \\ &= \frac{f(t)}{1 - F(t)} = \frac{\partial \log(1 - F(t))}{\partial t} = -\frac{\partial \log(S(t))}{\partial t} \end{aligned}$$

The cumulative hazard function is

$$H(t) = \int_0^t h(s)\mathsf{d}s = -\log(S(t))$$

and therefore

$$\begin{aligned} S(t) &= \exp(-H(t)) \\ f(t) &= h(t)\exp(-H(t)) \end{aligned}$$

Kaplan-Meier estimator

A step-function estimate of the survival curve
No assumptions about the distribution
Handles censoring naturally
Used to visualize and to compare time-to-event data of groups

$$\hat{S}(t) = \prod_{j:t_j \leq t} \left(1 - \frac{d_j}{n_j}\right)$$

t_j : event times (ordered)

d_j : number of events at time t_j

n_j : number "at risk" just before time t_j

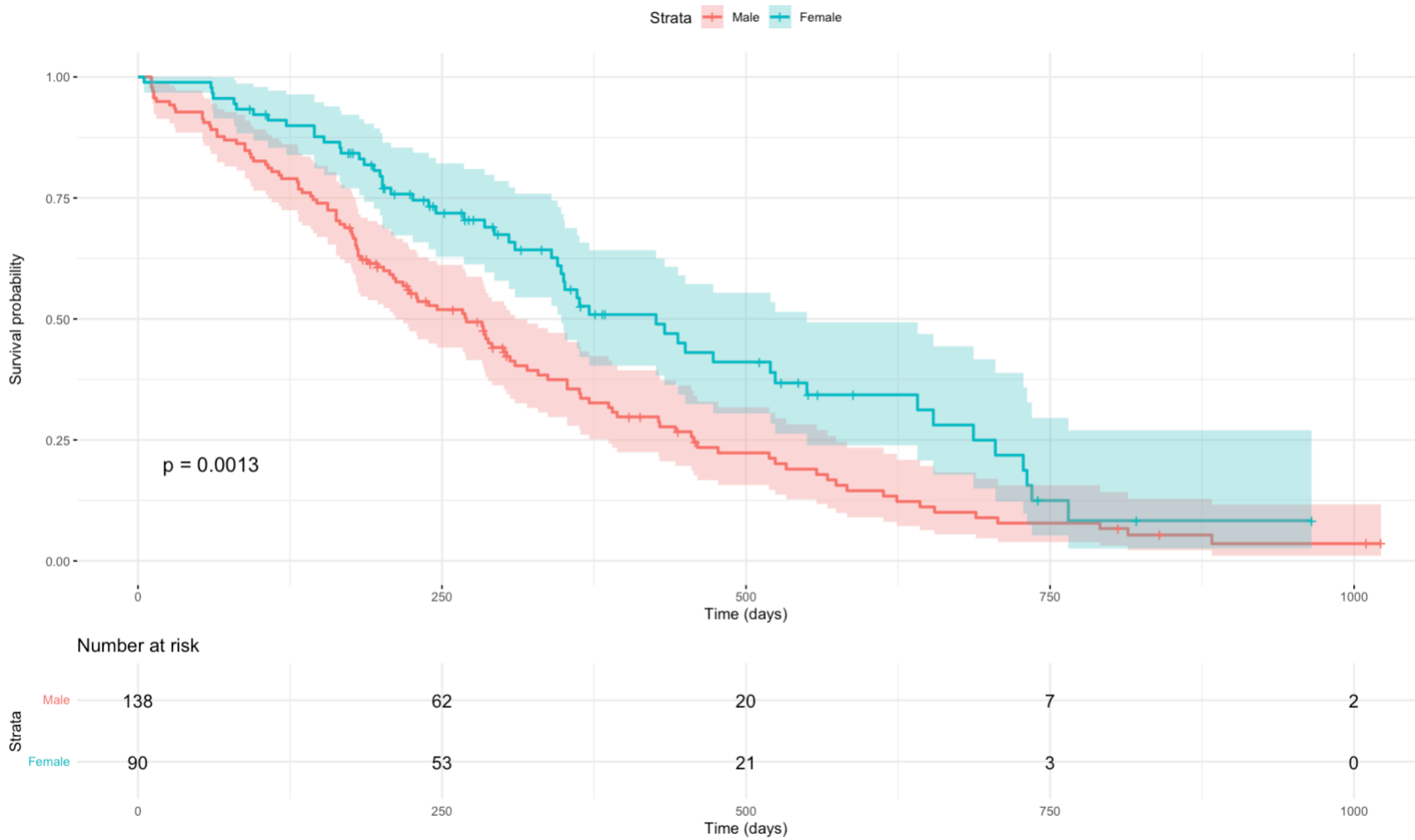
Sequential updating:

$$\hat{S}(t_0) = 1$$

$$\hat{S}(t_i) = \hat{S}(t_{i-1}) \times \left(1 - \frac{d_i}{n_i}\right)$$

R-Example: Kaplan-Meier estimator

lung dataset available from the survival package survival. The data contain subjects with advanced lung cancer from the North Central Cancer Treatment Group



Parametric survival models

There are many parametric survival models that assume a distribution for the survival time

- Exponential
- Weibull
- Log-normal
- Log-logistic
- Gompertz

We’ll only consider (semi-parametric) Cox proportional hazard model

Cox proportional hazard model

The most widely used survival model

Models the hazard ratio

explanatory variables

$$h(t \mid x) = h_0(t) \exp(x^T \beta)$$

baseline hazard coefficients

$\exp(\beta_j) > 1$: predictor has an increased hazard

$\exp(\beta_j) < 1$: predictors has a protective effect

No distribution assumption for the baseline hazard

Cox proportional hazard model

Cox-proportional hazard model $h(t \mid x) = h_0(t)\exp(x^T\beta)$
can be fitted if the hazard if the hazard ratio between any two individuals is constant over time

$$\frac{h(t \mid x_i)}{h(t \mid x_j)} = \text{Constant in } t$$

This means that
1) the effect of a covariate is multiplicative on the hazard
2) the effect does not change over time

R-example!

End of Lecture 11